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DEMOGRAPHY AND SETTLEMENT OF THE POST-SOVIET SPACE

Regional Population Trajectories, Urban Polarisation and Conflict Displacement in Belarus, Ukraine and Russia

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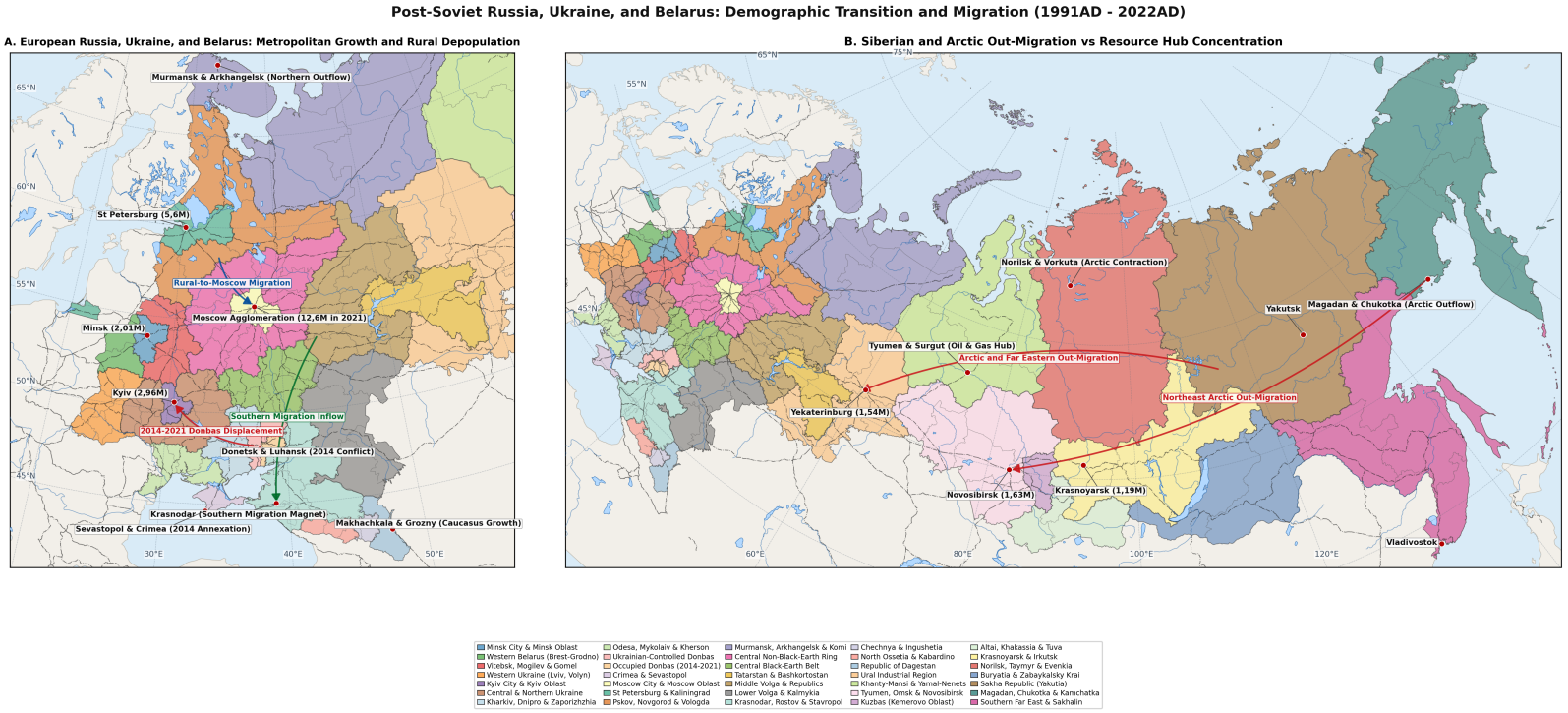


Figure 1. Boundaries of the 35 demographic regions across Belarus (Regions 1-3), Ukraine (Regions 4-11) and the Russian Federation (Regions 12-35), distinguishing metropolitan growth cores, Arctic and industrial contraction belts and the 2014-2021 conflict zones.

Abstract

Between the dissolution of the Soviet Union in December 1991 and the end of 2021, the combined population of Belarus, Ukraine and the Russian Federation (35 regions covering 17,97M km2) fell by 14,18M (-6,7%), from 212,26M to 198,08M. Belarus declined from 10,25M to 9,15M (-10,7%), the Russian Federation from 150,51M to 147,37M (-2,1%) and Ukraine within its 1991 borders from 51,5M to 41,56M (-19,3%). Beneath this overall contraction, below-replacement birth rates, surging working-age mortality ("the Russian and Ukrainian Cross") and the end of Soviet planning controls concentrated population growth into the capital regions of Moscow, Saint Petersburg, Minsk and Kyiv, the Southern Russian sunbelt, the West Siberian oil and gas okrugs and the North Caucasus republics, while emptying the Non-Black-Earth countryside, the Arctic periphery and the war-divided Donbas.

Background

The dissolution of the Soviet Union in December 1991 set in motion a thirty-year demographic transformation across Belarus, Ukraine and the Russian Federation. At the close of the Soviet era, the seven European and Northern Eurasian union republics held 224,55M inhabitants. The departure of Estonia (1,57M), Latvia (2,66M), Lithuania (3,7M) and Moldova (4,36M) removed 12,29M people from that total, leaving a 1991 baseline of 212,26M across the 35 regions examined here (17,97M km2): 10,25M in Belarus (Regions 1-3), 51,5M within the 1991 borders of Ukraine (Regions 4-11) and 150,51M in the Russian Federation (Regions 12-35). Return migration from Central Asia and the South Caucasus lifted the combined total briefly to a peak of 212,82M in 1995 [1][2][3][4].

Over the next quarter-century, this territory underwent a steep peacetime contraction alongside a sharp reordering of its urban and rural settlement hierarchy. By late 2021, on the eve of the full-scale Russian invasion of Ukraine in February 2022, the population across all 35 regions had fallen by 14,18M (-6,7% from 1991) to 198,08M. Belarus stood at 9,15M (-10,7%), the Russian Federation at 147,37M (-2,1%) and Ukraine within its 1991 borders at 41,56M (-19,3%), including 2,21M in Government-Controlled Donbas (Region 9), 2,47M in Pre-2022 Occupied Donbas (Region 10) and 2,44M in Crimea and Sevastopol (Region 11) [3][4][5][6].

Methodology

Population series for 1991-2021 rest on three national statistical systems that diverged in census frequency and territorial coverage after the breakup of the Union [7].

In the Russian Federation (Regions 12-35), following the 1989 All-Union Census (147,02M permanent residents, rising to 150,51M in 1991 and 151,39M in 1995), Rosstat conducted three national censuses: on 9 October 2002 (145,17M) [8], 14 October 2010 (142,86M) [9] and 1 October - 14 November 2021 (147,18M including Crimea and Sevastopol, or 147,37M across Regions 12-35) [5][10]. As Anatoly Vishnevsky, Sergei Zakharov, Evgueni Andreev and Nikita Mkrtchyan have shown, the 2002 and 2021 censuses relied heavily on police-registration records (*registratsiya*) in metropolitan Moscow (Region 12), Saint Petersburg (Region 13) and the North Caucasus (Regions 22-24), requiring cross-checks against civil vital-registration ledgers (ZAGS births and deaths), intercensal adjustments and municipal migration series [11][12][13][14].

In Ukraine (Regions 4-11), after the 1989 Soviet census (51,45M permanent residents, rising through CIS return migration to 52,2M in 1993), independent Ukraine held only one door-to-door census, on 5 December 2001 (48,46M) [15]. Following the Russian annexation of Crimea and Sevastopol in March 2014 (Region 11) and the outbreak of war in the Donbas in April 2014 - which divided Donetsk and Luhansk oblasts along a 427 km Line of Contact between Government-Controlled Donbas (Region 9) and Pre-2022 Occupied Donbas (Region 10) - Ukrainian series were maintained through annual Ukrstat vital-registration balances [3], Ptoukha Institute reconstructions and the December 2019 Electronic Population Assessment (37,29M on government-controlled territory, or 41,56M across Regions 4-11 in 2021) [6][16][17][18].

In Belarus (Regions 1-3), Belstat conducted decennial censuses on 16 February 1999 (10,04M) [19], 14 October 2009 (9,5M) [20] and 4-30 October 2019 (9,41M, declining to 9,15M across Regions 1-3 by late 2021 under COVID-19 excess mortality and post-2020 emigration) [4][21][22].

To trace these divergent demographic regimes, this study divides Belarus (Regions 1-3), Ukraine (Regions 4-11) and the Russian Federation (Regions 12-35) into 35 regions. In the North Caucasus, this division separates Chechnya and Ingushetia (Region 22, shaped by the Chechen Wars of 1994-2009, the Nazran displacement surge and post-war recovery), North Ossetia, Kabardino-Balkaria and Karachay-Cherkessia

(Region 23, site of the 1992 Prigorodny conflict) and Dagestan (Region 24, marked by sustained natural increase and the Makhachkala-Kaspiysk urban expansion). In Ukraine, it separates Government-Controlled Donbas (Region 9), Pre-2022 Occupied Donbas (Region 10) and Crimea and Sevastopol (Region 11). Across Asian Russia, ten regions (Regions 26-35) distinguish the Trans-Siberian and Pacific corridors (Regions 27, 28, 30, 32 and 35) from the southern Siberian republics of Altai, Khakassia and Tuva (Region 29) and the northern hydrocarbon and polar zones of Yugra and Yamal (Region 26), Norilsk, Taymyr and Evenkia (Region 31), Yakutia (Region 33) and Magadan, Chukotka and Kamchatka (Region 34).

Results

Regional Population Trajectories

Figure 2 compares ten representative regional trajectories with the indexed national curves (1991 = 100), while Table 1 presents the benchmark series (1991, 1995, 2000, 2005, 2010, 2013, 2015 and 2021) across all 35 regions.

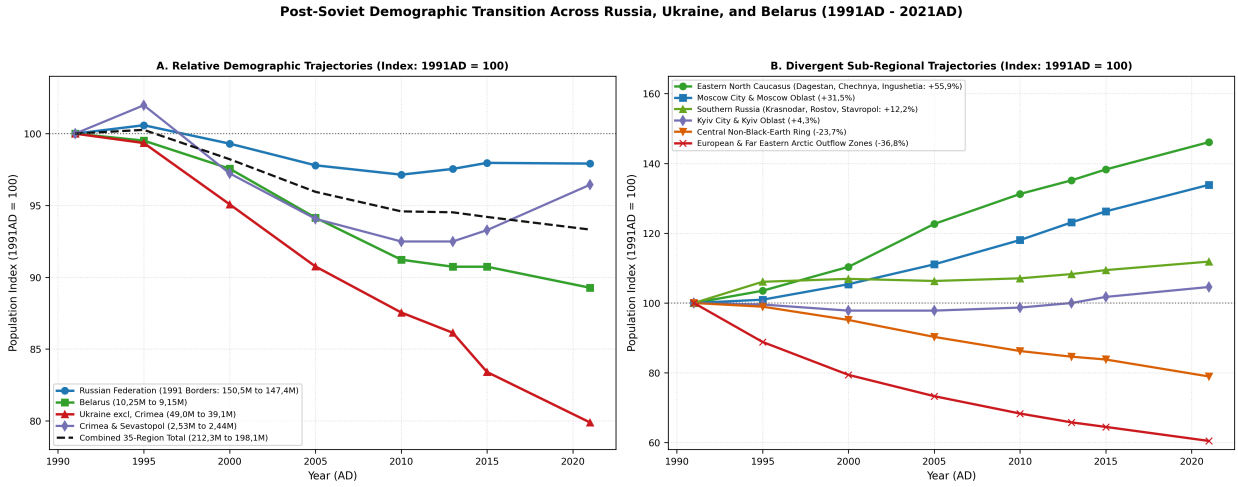


Figure 2. Indexed population curves (1991 = 100) for Belarus, Ukraine and the Russian Federation compared with ten contrasting growth, industrial and conflict regions from 1991 to 2021.

Region	1991	1995	2000	2005	2010	2013	2015	2021	Change
1. Minsk City & Minsk Oblast Metropolitan Magnet	3,4M	3,39M	3,36M	3,32M	3,34M	3,38M	3,42M	3,48M	+2,4%
2. Western Belarus: Brest & Grodno Oblasts	2,68M	2,69M	2,65M	2,56M	2,47M	2,44M	2,43M	2,36M	-11,9%

3. Eastern & Southeastern Belarus: Vitebsk, Mogilev & Gomel Oblasts	4,17M	4,12M	3,99M	3,77M	3,54M	3,48M	3,45M	3,31M	-20,6%
4. Western Ukraine	10,25M	10,28M	10,05M	9,78M	9,62M	9,58M	9,56M	9,32M	-9,1%
5. Kyiv City & Kyiv Metropolitan Oblast	4,58M	4,56M	4,48M	4,48M	4,52M	4,58M	4,66M	4,79M	+4,6%
6. Central & Northern Ukraine Depopulation Belt	12,12M	11,99M	11,39M	10,69M	10,08M	9,77M	9,66M	9,03M	-25,5%
7. Kharkiv, Dnipro-petrovsk & Zaporizhzhia Industrial East	9,1M	9,02M	8,58M	8,12M	7,78M	7,62M	7,65M	7,25M	-20,3%
8. Ukrainian Black Sea Littoral	4,7M	4,68M	4,48M	4,32M	4,22M	4,18M	4,16M	4,05M	-13,8%
9. Government-Controlled Donbas	3,62M	3,58M	3,35M	3,12M	2,95M	2,86M	2,45M	2,21M	-39%
10. Occupied Donbas / ORDLO	4,6M	4,54M	4,23M	3,93M	3,7M	3,59M	2,7M	2,47M	-46,3%
11. Crimea & Sevastopol	2,53M	2,58M	2,46M	2,38M	2,34M	2,34M	2,36M	2,44M	-3,6%
12. Moscow City & Moscow Oblast Post-Soviet Hyper-Centralisation Core	15,8M	15,95M	16,65M	17,55M	18,65M	19,45M	19,95M	21,15M	+33,9%
13. Saint Petersburg, Leningrad Oblast & Kaliningrad Exclave	7,62M	7,55M	7,42M	7,38M	7,52M	7,75M	7,95M	8,42M	+10,5%

14. Depopulating Northwest: Pskov, Novgorod, Vologda & Republic of Karelia	3,72M	3,66M	3,48M	3,26M	3,08M	3,01M	2,96M	2,8M	-24,7%
15. European Arctic Exodus Zone: Murmansk, Arkhangelsk, Komi & Nenets AO	4,32M	3,99M	3,63M	3,38M	3,15M	3,03M	2,97M	2,76M	-36,1%
16. Central Non-Black-Earth Mortality & Shrinkage Ring	12,35M	12,22M	11,75M	11,15M	10,65M	10,45M	10,35M	9,75M	-21,1%
17. Central Black-Earth Belt	7,75M	7,98M	7,85M	7,58M	7,38M	7,32M	7,3M	7,02M	-9,4%
18. Republics of Tatarstan & Bashkor-tostan	7,58M	7,85M	7,89M	7,84M	7,86M	7,91M	7,95M	7,95M	+4,9%
19. Middle Volga Oblasts	15,4M	15,68M	15,35M	14,75M	14,25M	14,05M	13,95M	13,31M	-13,5%
20. Lower Volga	7,12M	7,37M	7,27M	7,04M	6,86M	6,81M	6,77M	6,44M	-9,6%
21. Southern Russian Sunbelt Magnet	13,15M	13,95M	14,06M	13,98M	14,08M	14,24M	14,39M	14,71M	+11,9%
22. Chechnya & Ingushetia	1,32M	1,18M	1,12M	1,52M	1,69M	1,79M	1,86M	2,04M	+54,5%
23. North Ossetia	1,92M	1,98M	2,08M	2,11M	2,12M	2,11M	2,11M	2,11M	+9,9%
24. Republic of Dagestan	1,88M	2,14M	2,45M	2,65M	2,91M	3,02M	3,11M	3,33M	+77,1%
25. Ural Industrial Belt	16,32M	16,47M	16,29M	15,87M	15,61M	15,54M	15,5M	15,7M	-3,8%
26. Khanty-Mansi	1,82M	1,81M	1,88M	1,99M	2,06M	2,13M	2,16M	2,26M	+23,9%
27. West Siberian Trans-Siberian Belt	11,15M	11,22M	11,03M	10,72M	10,54M	10,55M	10,6M	10,49M	-5,9%

28. Kuzbass Coal-Metallurgical Basin	3,18M	3,1M	2,98M	2,86M	2,76M	2,74M	2,72M	2,6M	-18,2%
29. Southern Siberian Republics: Altai, Khakassia & Tuva	1,09M	1,08M	1,06M	1,04M	1,04M	1,05M	1,06M	1,08M	-0,5%
30. East Siberian Trans-Siberian & Angara-Yenisei Corridor	6,03M	5,94M	5,71M	5,46M	5,32M	5,3M	5,29M	5,17M	-14,3%
31. Norilsk, Taymyr & Evenkia Polar De-Population Zone	380k	330k	290k	260k	235k	225k	220k	220k	-42,1%
32. Republic of Buryatia	2,41M	2,38M	2,29M	2,19M	2,08M	2,06M	2,05M	2M	-17,2%
33. Sakha Republic	1,11M	1,03M	960k	950k	958k	955k	960k	985k	-11,3%
34. Extreme Northeast Arctic Exodus Zone	1,03M	770k	630k	560k	530k	515k	505k	485k	-52,9%
35. Southern Far East & Pacific Corridor	6,06M	5,76M	5,34M	5,1M	4,88M	4,8M	4,76M	4,58M	-24,3%
Sub-Total: Republic of Belarus (Regions 1-3)	10,25M	10,2M	10M	9,65M	9,35M	9,3M	9,3M	9,15M	-10,7%
Sub-Total: Ukraine [1991 Borders] (Regions 4-11)	51,5M	51,23M	49,02M	46,82M	45,21M	44,52M	43,2M	41,56M	-19,3%

Sub-Total: Russian Federation [1991 Borders] (Regions 12-35)	150,51M	151,39M	149,46M	147,19M	146,21M	146,81M	147,44M	147,37M	-2,1%
Total Post-Soviet Space (Regions 1-35)	212,26M	212,82M	208,48M	203,66M	200,77M	200,63M	199,94M	198,08M	-6,7%

Table 1. Benchmark population estimates and net percentage change across the 35 regions of Belarus, Ukraine and the Russian Federation between 1991 and 2021.

As Table 1 shows, the 35-region division highlights six regional patterns between 1991 and 2021 [14][23][24].

Across the three Belarusian regions (Regions 1-3, declining overall from 10,25M to 9,15M, or -10,7%), Region 1 (Minsk City and Minsk Oblast) expanded by +2,4% (3,4M to 3,48M) as Minsk city (1,61M to 2,01M) absorbed young migrants from across the republic. By contrast, Region 2 (Western Belarus: Brest and Grodno oblasts) contracted by -11,9% (2,68M to 2,36M), while Region 3 (Eastern and Southeastern Belarus: Vitebsk, Mogilev and Gomel oblasts) fell by -20,6% (4,17M to 3,31M) under post-Chernobyl resettlement in Gomel and Mogilev and rural ageing across Vitebsk [21][22].

Across the eight Ukrainian regions (Regions 4-11, falling from 51,5M to 41,56M, or -19,3%), only Region 5 (Kyiv City and Kyiv Oblast) recorded net growth (4,58M to 4,79M, +4,6%). Region 4 (Western Ukraine: Galicia, Volhynia, Bukovina and Zakarpattia) fell by -9,1% (10,25M to 9,32M) as higher rural fertility partially cushioned labour emigration to the European Union; Region 8 (Ukrainian Black Sea Littoral: Odesa, Mykolaiv and Kherson) declined by -13,8% (4,7M to 4,05M); Region 7 (Kharkiv, Dnipropetrovsk and Zaporizhzhia) contracted by -20,3% (9,1M to 7,25M, briefly rising from 7,62M in 2013 to 7,65M in 2015 as the primary reception belt for displaced residents from the Donbas); and Region 6 (Central and Northern Ukraine) fell by -25,5% (12,12M to 9,03M) amid severe rural ageing, with crude death rates exceeding 19-22 per 1k in Chernihiv and Sumy. In the conflict and annexation zones, Region 11 (Crimea and Sevastopol, annexed in March 2014) stood at 2,44M in 2021 (-3,6% relative to 1991, rising from 2,34M in 2013 through Russian military and civilian inflows), whereas the partitioned Donbas coal basin suffered steep depopulation along the Line of Contact: Region 9 (Government-Controlled Donbas: Mariupol, Kramatorsk and Sievierodonetsk) fell by -39% (from 3,62M in 1991 to 2,86M in 2013 and 2,21M in 2021) and Region 10 (Pre-2022 Occupied Donbas: Donetsk-Makiivka-Horlivka and Luhansk-Alchevsk) dropped by -46,3% (from 4,6M in 1991 to 3,59M in 2013 and 2,47M in 2021) [16][17][25].

Five regions in the Russian Federation outside the Caucasus acted as metropolitan, sunbelt, hydrocarbon and petrochemical magnets (Regions 12, 13, 18, 21 and 26). Region 12 (Moscow City and Moscow Oblast) surged by +33,9% (15,8M to 21,15M, +5,35M), Region 26 (Khanty-Mansi/Yugra and Yamalo-Nenets autonomous okrugs) grew by +23,9% (1,82M to 2,25M), Region 21 (Southern Russian Sunbelt: Krasnodar, Rostov, Stavropol and Adygea) expanded by +11,9% (13,15M to 14,71M, +1,56M), Region 13 (Saint Petersburg, Leningrad Oblast and Kaliningrad) rose by +10,5% (7,62M to 8,42M) and Region 18 (Tatarstan and Bashkortostan) increased by +4,9% (7,58M to 7,95M) [24][26][27].

In the North Caucasus republics (Regions 22, 23 and 24), Region 24 (Dagestan) recorded the fastest sustained expansion in the post-Soviet space at +77,1% (1,88M to 3,33M), propelled by above-replacement fertility and the growth of the Makhachkala-Kaspiysk conurbation [28]. Region 22 (Chechnya and Ingushetia) fell from 1,32M in 1991 to 1,12M in 2000 during the First and Second Chechen Wars and the permanent departure of ethnic Russians from Grozny, before rebounding through Vainakh natural increase (total fertility rate ~2,5-2,8), the expansion of Nazran and federal reconstruction to 2,04M in 2021 (+54,5%) [29][30]. Between them, Region 23 (North Ossetia-Alania, Kabardino-Balkaria and Karachay-Cherkessia) experienced both the 1992 Prigorodny conflict and refugee inflows from South Ossetia, growing by +9,9% (1,92M to 2,11M).

Across the polar and Arctic zones (Regions 15, 31, 33 and 34), disaggregating the Russian North shows the scale of the post-Soviet *Severnyy ottok* ("Northern Outflow"): Region 34 (Magadan, Chukotka and Kamchatka) lost more than half its inhabitants at -52,9% (1,03M to 485k); Region 31 (Norilsk, Taymyr and Evenkia) fell by -42,1% (380k to 220k); and Region 15 (Murmansk, Arkhangelsk, Komi and Nenets AO) declined by -36,1% (4,32M to 2,76M). By contrast, Region 33 (Sakha Republic/Yakutia) contracted by only -11,3% (1,11M to 985k, recovering from a 2005 trough of 950k) because higher indigenous Sakha fertility and the rapid expansion of Yakutsk (187k to 355k, +89,8%) partly offset the emptying of northern Russian mining settlements [31][32][33].

Along the Trans-Siberian corridors, southern Siberian republics and European interior belts (Regions 14, 16-17, 19-20, 25, 27-30, 32 and 35), ageing and rural depopulation in European Russia reduced Region 14 (Pskov, Novgorod, Vologda and Karelia) by -24,7% (3,72M to 2,8M) and Region 16 (Central Non-Black-Earth Ring) by -21,1% (12,35M to 9,75M), alongside losses in the Middle Volga and Finno-Ugric republics (Region 19: -13,5%), the Lower Volga and Kalmykia (Region 20: -9,6%), the Central Black-Earth belt (Region 17: -9,4%) and the Urals (Region 25: -3,8%). In southern Siberia, Region 29 (Altai, Khakassia and Tuva) held nearly steady at -0,5% (1,09M to 1,08M) as high Tuvan and Altai fertility (TFR ~2,5-3) balanced ethnic Russian out-migration, while Region 27 (West Siberian Corridor) fell by only -5,9% (11,15M to 10,49M) thanks to the expansion of Tyumen (+77,6%) and Novosibirsk (+13,7%). Coal-mine closures drove Region 28 (Kuzbass/Kemerovo Oblast) down by -18,2% (3,18M to 2,6M), while westward out-migration (*Zapadnyy dreyf*) reduced Region 30 (Krasnoyarsk and Irkutsk) by -14,3% (6,03M to 5,17M), Region 32 (Buryatia and Zabaykalsky Krai) by -17,2% (2,41M to 2M) and Region 35 (Southern Far East: Primorye, Khabarovsk, Amur, Jewish AO and Sakhalin) by -24,3% (6,06M to 4,58M) [34][35][36].

Mortality Crisis and Migration

The defining demographic shock of the immediate post-Soviet transition (1991-2006) was the simultaneous collapse of period fertility and surge in working-age mortality across Russia, Ukraine and Belarus - a scissors-like divergence of vital rates known as the "Russian Cross" (*Russkiy Krest*) and "Ukrainian Cross" [11][12][37][38]. Figure 3 plots the crude birth rate (CBR) and crude death rate (CDR) of Russia and Ukraine from 1990 to 2021 alongside net percentage population change across all 35 regions, while Table 2 summarises decadal vital rates, life expectancy at birth (e_0), cumulative natural decrease and net migration balances.

Country and Period	CBR/CDR (per 1k)	TFR	Male/ Female e0	Natural Balance	Net Migration
Russian Federation: Late Soviet Baseline (1989-1991)	13,4/11,2	1,89	63,8/74,3	+680k	+490k

Russian Federation: Transition and Hyperinflation Crisis (1992-1999)	9,3/14,5	1,16 (1999 trough)	57,6/71,2 (1994)	-5,82M	+3,65M (CIS return wave)
Russian Federation: Second Mortality Wave (2000-2006)	9,8/15,8	1,3	58,9/72 (2003)	-5,96M	+1,18M
Russian Federation: Pronatalist Recovery and Parity Window (2007-2016)	12,6/13,6	1,76 (2015 peak)	66,5/77,1 (2016)	-1,45M (+83k in 2013-2015)	+2,65M
Russian Federation: Structural Decline and COVID-19 Shock (2017-2021)	10,4/13,9 (16,7 in 2021)	1,5	65,5/74,5 (2021)	-2,52M (-1,04M in 2021)	+980k
Ukraine [1991 Borders]: Late Soviet Baseline (1989-1991)	12,6/12,1	1,85	65,6/74,9	+95k	+240k
Ukraine: Hyperinflation, Fertility Collapse and Emigration (1992-2001)	9,1/14,8	1,08 (2001 world low)	61,2/72,8 (1996)	-2,98M	-1,15M
Ukraine: Post-2001 Depopulation and Stabilisation (2002-2013)	10,1/15,6	1,46	62,3/74,1	-3,12M	-380k
Ukraine: Donbas War, Emigration and COVID-19 Shock (2014-2021)	9,2/15,4 (18,5 in 2021)	1,22 (1,16 in 2021)	65,2/74,4 (2021)	-2,18M (-442k in 2021)	-1,45M (Donbas/EU outflow)
Republic of Belarus: Late Soviet Baseline (1989-1991)	13,9/10,7	1,91	66,3/75,6	+92k	+35k
Republic of Belarus: Post-Soviet Depopulation and Rural Ageing (1992-2005)	9,6/13,8	1,25 (2003 trough)	62,2/74,1 (2002)	-580k	+110k
Republic of Belarus: Stabilisation and COVID-19 Shock (2006-2021)	11,1/13,2 (16,4 in 2021)	1,45 (1,72 in 2015)	67,3/77,7	-480k	-95k

Table 2. Crude vital rates, period fertility, life expectancy at birth, natural balance and net international migration in Russia, Ukraine and Belarus by demographic phase, 1989-2021.

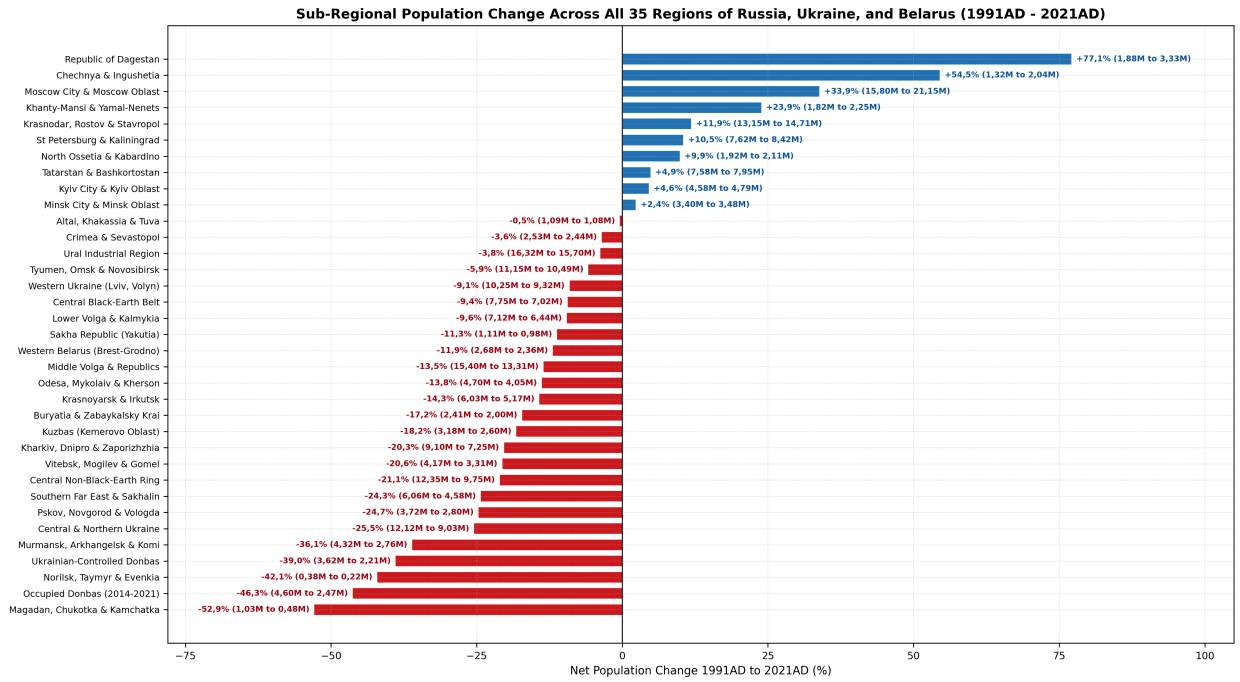


Figure 3. Annual crude birth and death rates (per 1k population) in Russia and Ukraine from 1990 to 2021, paired with cumulative regional population change across all 35 regions.

Hyperinflation and Fertility Collapse

Before the breakup of the Soviet Union, period fertility across all three East Slavic republics stood near replacement (TFR ~1,85-2,01 in 1988-1990), supported by early marriage (mean age at first birth ~22,5 years) and state housing allocation [11][12]. Price liberalisation in January 1992, hyperinflation - with consumer prices rising more than 25-fold in Russia in 1992 and 100-fold in Ukraine in 1993 - a 40-60% drop in real wages, widespread wage arrears and the closure of enterprise childcare facilities triggered an immediate fall in second and higher-order births [18][39].

In the Russian Federation (Regions 12-35), live births fell by nearly half, from 2,16M (14,6 per 1k) in 1989 to 1,22M (8,3 per 1k) in 1999, driving the total fertility rate down to a trough of 1,16 in 1999 (0,92 in Saint Petersburg, Region 13, and 1,01 in Moscow, Region 12) [2][12].

In Ukraine (Regions 4-11), live births fell from 691k (13,4 per 1k) in 1989 to 376k (7,7 per 1k) in 2001, with the TFR reaching 1,08 in 2001 - then the lowest national fertility rate recorded in the world (0,89 in Kyiv, Region 5, and 0,91-0,95 across the Donbas, Regions 9-10, and the industrial east, Region 7) [16][18].

In Belarus (Regions 1-3), live births declined from 153k (15 per 1k) in 1989 to 88,5k (8,9 per 1k) in 2002-2003, with the TFR bottoming at 1,25 [20][21].

As Sergei Zakharov has shown, this fertility trough reflected the intersection of acute economic hardship with the Second Demographic Transition: cohorts born in the 1970s and 1980s moved away from the Soviet pattern of universal childbearing at ages 20-23 toward delayed marriage and later motherhood, even as older cohorts curtailed progression beyond a first child [12][40].

Working-Age Male Mortality

While birth rates collapsed, mortality surged across all three countries between 1992 and 1994, eased briefly in 1995-1998 and rose again after the August 1998 rouble default to a second peak in 2003 [13][37][41]. Following the repeal of the state alcohol monopoly in 1992, the spread of cheap untaxed vodka and industrial alcohol surrogates (*samogon*, colognes and medicinal tinctures), combined with economic dislocation and deteriorating cardiovascular care, produced a sharp rise in premature deaths among men aged 25-59 [42][43][44].

Between 1991 and 1994, annual deaths in Russia rose from 1,69M (11,4 per 1k) to 2,3M (15,7 per 1k, exceeding 21-24 per 1k in Pskov and Novgorod in Region 14 and in Tver, Tula and Ivanovo in Region 16). Russian male life expectancy fell by 7,3 years in five years to 57,6 years in 1994, compared with 71,2 years for women - a gender gap of 13,6 years [13][37].

After the 1998 financial crash, Russian deaths reached 2,37M (16,4 per 1k) in 2003 (male e_0 = 58,9 years). Epidemiological studies, notably the Izhevsk Family Study in Region 19 led by David Leon, Vladimir Shkolnikov and Evgueni Andreev, showed that hazardous binge drinking of spirits and non-beverage alcohols accounted for 43% of all deaths among working-age Russian men [38][42][43].

In Ukraine, annual deaths rose from 630k (12,1 per 1k) in 1990 to 792k (15,2 per 1k) in 1995-1996 and 765k (16,2 per 1k) in 2005-2006 (reaching 20-23 per 1k in Chernihiv and Sumy oblasts in Region 6), with male life expectancy falling to 61,2 years [38][45]. Between 1992 and 2006 inclusive, cumulative excess deaths over births reached -11,78M in the Russian Federation, -5,62M in Ukraine and -750k in Belarus - a combined natural deficit of 18,15M lives over fifteen years [2][3][20].

Repatriation and Emigration

Why did the population of the Russian Federation decline by only -2,1% between 1991 and 2021 (150,51M to 147,37M) despite more than 14M in cumulative natural decrease, whereas Ukraine's population fell by -19,3% (51,5M to 41,56M)? The explanation lies in three international migration streams [32][46][47].

Between 1992 and 2002, repatriation from other former Soviet republics brought Russia a net gain of +4,83M permanent settlers (peaking at +845k net in 1994 alone and lifting Russia's regional total from 150,51M in 1991 to 151,39M in 1995) from Kazakhstan (+1,85M), Central Asia (+1,42M), the South Caucasus (+820k) and the Baltic states (+390k) [46][48]. These migrants settled heavily in the Southern Sunbelt (Region 21: Krasnodar, Stavropol and Rostov, rising from 13,15M to 13,95M in 1995), the Central Black-Earth belt (Region 17: Belgorod and Voronezh, 7,75M to 7,98M), the Middle Volga (Region 19: Samara, 15,4M to 15,68M), the Lower Volga (Region 20: Saratov and Volgograd, 7,12M to 7,37M) and the West Siberian Corridor (Region 27: Novosibirsk, Omsk and Altai Krai, 11,15M to 11,22M), offsetting roughly 55% of Russia's natural decline during the 1990s.

At the same time, the lifting of Soviet exit restrictions between 1989 and 2005 led to the permanent emigration of more than 1,15M post-Soviet Jews (*Aliyah* to Israel alongside movement to the United States and Germany) and more than 1,85M ethnic Germans (*Spätaussiedler* to Germany) [32][46]. Across Ukraine (Regions 5-8), the Jewish population fell from 486k in 1989 to 103k at the 2001 census, above all in Kyiv (Region 5), Odesa (Region 8), Kharkiv and Dnipro (Region 7); in Belarus (Regions 1 and 3), from 112k in 1989 to 28k in 1999; and in Russia, from 551k in 1989 to 230k in 2002 and 83k in 2021, including the departure of more than half the Jewish population of Birobidzhan in the Jewish Autonomous Oblast (Region 35) [1][8][15]. Over 450k ethnic Germans emigrated directly to Germany from the German national districts of Omsk Oblast (Azovo) and Altai Krai (Halbstadt) in Region 27, from Saratov and Volgograd (Region 20) and from the Urals (Region 25), while more than 900k Germans from Kazakhstan transited through West Siberia and the Volga [46].

Ukraine initially recorded a net migration gain of +540k in 1991-1993, including roughly 250k Crimean Tatars returning from deportation in Uzbekistan to Crimea (Region 11, rising from 2,53M to 2,58M in 1995) [15][16]. From 1994 onward, however, deeper economic contraction turned Ukraine into a country of net emigration, losing more than 1,4M residents in 1994-2001 and 2M-3M labour migrants from Western Ukraine (Region 4) and Central Ukraine (Region 6) during the 2000s and 2010s [18][49][50].

Northern Exodus and Urban Polarisation

Alongside natural decrease, the thirty years between 1991 and 2021 brought a sharp spatial polarisation of settlement across the 35 regions of Russia, Ukraine and Belarus [14][24][26][51]. Soviet central planning had placed tens of millions of workers in high-latitude Arctic mining towns, Siberian timber and defence settlements and single-enterprise industrial towns (*monogoroda*) through northern wage supplements (*severnye nadbavki*) and subsidised supply lines (*severnyy zavoz*) [31][34]. When price liberalisation in 1992 exposed the true transport and heating costs of high-latitude and single-factory settlements, two migratory currents reshaped the map: an east-to-west and north-to-south internal exodus (*Zapadnyy dreyf*, or "Western Drift", and *Severnyy ottok*, or "Northern Outflow") from the Northeast Arctic (Region 34), Norilsk, Taymyr and Evenkia (Region 31), the European Arctic (Region 15) and the Southern Far East (Region 35); and centripetal concentration into Moscow (Region 12), Saint Petersburg and Kaliningrad (Region 13), the Southern Sunbelt (Region 21), the Yugra and Yamal hydrocarbon zone (Region 26), Minsk (Region 1) and Kyiv (Region 5) [33][52][53]. Figure 4 contrasts expanding metropolitan and sunbelt poles with contracting Arctic, coal-mining and conflict-affected cities from 1989 to 2021, while Table 3 summarises representative regional and urban shifts.

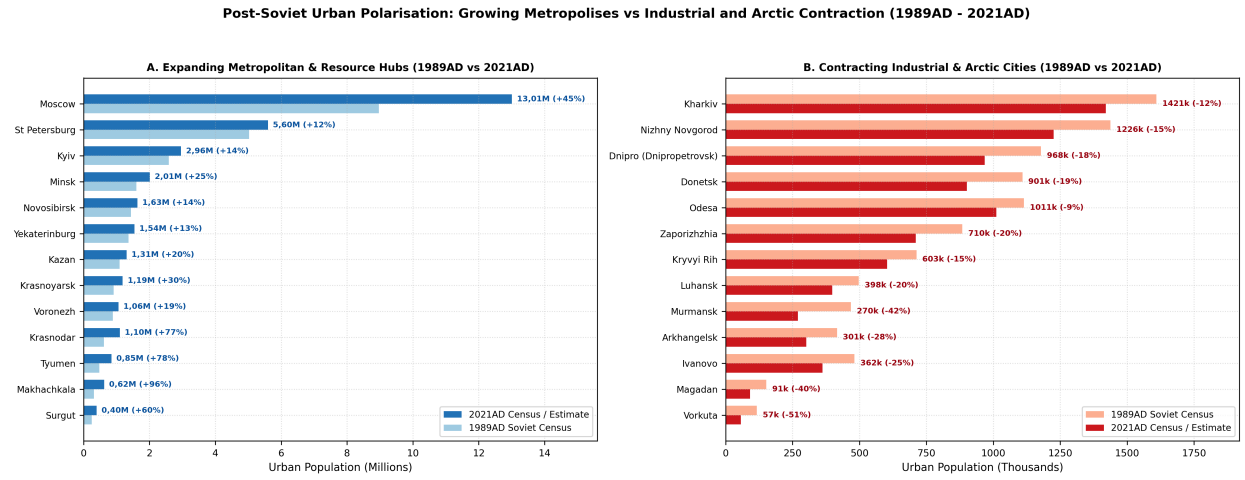


Figure 4. Contrasting city-level population trajectories between expanding metropolitan, sunbelt and hydrocarbon centres and shrinking Arctic and coal-basin towns, 1989-2021.

Region and Representative Settlements	1989/1991	2001/2002	2010	2021	Change
Northeast Arctic Exodus: Magadan, Chukotka and Kamchatka (Region 34)	1,03M (Chukotka 164k, Magadan 392k)	630k (Chukotka 54k, Magadan 183k)	530k (Chukotka 51k, Magadan 157k)	485k (Chukotka 50k, Magadan 136k)	-52,9% (Kadykchan abandoned)
Polar Depopulation: Norilsk, Taymyr and Evenkia (Region 31)	380k (Norilsk aggl. 265k)	290k (Norilsk 215k)	235k (Norilsk 175k)	220k (Norilsk 175k, Igarka 3,6k)	-42,1% (Igarka -80,7%)
European Arctic Exodus: Murmansk, Arkhangelsk, Komi and Nenets AO (Region 15)	4,32M (Murmansk 468k, Vorkuta 116k)	3,63M (Murmansk 336k, Vorkuta 85k)	3,15M (Murmansk 307k, Vorkuta 71k)	2,76M (Murmansk 270k, Vorkuta 57k)	-36,1% (Vorkuta -50,9%)
Southern Far East Corridor: Primorye, Khabarovsk, Amur and Sakhalin (Region 35)	6,06M (Vladivostok 634k)	5,34M (Vladivostok 595k)	4,88M (Vladivostok 592k)	4,58M (Vladivostok 601k)	-24,3% (Amur -27,5%)
Yakutia Urbanisation vs Mining Shrinkage (Region 33)	1,11M (Yakutsk 187k, Tiksi 11,6k)	960k (Yakutsk 211k, Tiksi 5,9k)	958k (Yakutsk 270k, Tiksi 5,1k)	985k (Yakutsk 355k, Tiksi 4,2k)	-11,3% (Yakutsk +89,8%)
Kuzbass Coal Restructuring: Kemerovo, Novokuznetsk and Prokopyevsk (Region 28)	3,18M (Prokopyevsk 274k)	2,98M (Prokopyevsk 225k)	2,76M (Prokopyevsk 210k)	2,6M (Prokopyevsk 178k)	-18,2% (Prokopyevsk -35%)
Ural Industrial Monogoroda: Nizhny Tagil, Zlatoust and Berezniki (Region 25)	16,32M (N. Tagil 440k, Berezniki 201k)	16,29M (N. Tagil 390k, Berezniki 173k)	15,61M (N. Tagil 362k, Berezniki 156k)	15,7M (N. Tagil 339k, Berezniki 138k)	-3,8% (Yekaterinburg +13%)
Central Non-Black-Earth Ring: Ivanovo, Tula and Tver (Region 16)	12,35M (Ivanovo 481k, Tula 540k)	11,75M (Ivanovo 432k, Tula 481k)	10,65M (Ivanovo 408k, Tula 501k)	9,75M (Ivanovo 362k, Tula 473k)	-21,1% (Ivanovo -24,7%)

Metropolitan Growth: Moscow City and Oblast (Region 12)	15,8M (Moscow city 8,97M)	17M (Moscow city 10,38M)	18,65M (Moscow city 11,5M)	21,15M (Moscow city 13,01M)	+33,9% (Moscow +45,1%)
Moscow Ring Boomtowns: Balashikha, Khimki, Mytishchi and Krasnogorsk (Region 12)	539k combined	586k combined	765k combined	1,22M (Balashikha 521k)	+126,5% (Balashikha +285,9%)
Baltic and Northwest Core: Saint Petersburg, Leningrad and Kaliningrad (Region 13)	7,62M (StP 5,02M)	7,42M (StP 4,66M)	7,52M (StP 4,88M)	8,42M (StP 5,6M; Murino 89k)	+10,5% (StP +11,5%)
Southern Sunbelt: Krasnodar, Sochi, Rostov and Stavropol (Region 21)	13,15M (Krasnodar 621k)	14,06M (Krasnodar 646k)	14,08M (Krasnodar 745k)	14,71M (Krasnodar 1,1M)	+11,9% (Krasnodar +77%)
Arctic Hydrocarbon Belt: Surgut, Nizhnevar-tovsk and Novy Urengoy (Region 26)	1,82M (Surgut 248k)	1,88M (Surgut 285k)	2,06M (Surgut 307k)	2,25M (Surgut 396k, Kh.-Mansiysk 107k)	+23,9% (Surgut +59,7%)
West Siberian Growth Poles: Tyumen and Novosibirsk (Region 27)	11,15M (Tyumen 477k, Novosib. 1,44M)	11,03M (Tyumen 511k, Novosib. 1,43M)	10,54M (Tyumen 582k, Novosib. 1,47M)	10,49M (Tyumen 847k, Novosib. 1,63M)	-5,9% (Tyumen +77,6%)
Caucasus and South Siberian Capitals: Makhachkala (Reg. 24) and Kyzyl (Reg. 29)	1,88M Reg. 24 (Makhachkala 318k; Kyzyl 87k)	2,45M Reg. 24 (Makhachkala 462k; Kyzyl 104k)	2,91M Reg. 24 (Makhachkala 572k; Kyzyl 110k)	3,33M Reg. 24 (Makhachkala 623k; Kyzyl 125k)	+77,1% Reg. 24 (Makhachkala +95,9%)

East Slavic Capitals: Kyiv Metropolis (Region 5) and Minsk Region (Region 1)	7,98M (4,58M + 3,4M)	7,84M (4,48M + 3,36M)	7,86M (4,52M + 3,34M)	8,27M (4,79M + 3,48M)	+3,6% (Minsk city +25,1%)

Table 3. Population shifts across major depopulation belts and expanding metropolitan, sunbelt and hydrocarbon growth poles between 1989 and 2021.

The Northern Outflow and Siberian Corridors

Disaggregating Asian Russia and the European North into eleven regions (Region 15 and Regions 26-35) shows why treating Siberia and the Far East as a single block obscures three distinct demographic patterns [31][32][33][34].

Across the northeast and polar depopulation zones (Regions 34, 31 and 15), Region 34 (Magadan, Chukotka and Kamchatka) saw its population fall by -52,9% from 1,03M in 1991 to 485k in 2021: Chukotka Autonomous Okrug lost -69,5% (164k to 50k), Magadan Oblast lost -65,3% (392k to 136k) and Kamchatka Krai declined from 472k to 292k (-38,1%, with Petropavlovsk-Kamchatsky falling from 269k to 165k). Dozens of Kolyma and Chukotka mining settlements - including Iultin, Valkumey and the coal town of Kadykchan (10,3k in 1989, abandoned after its heating plant exploded in 1996) - were closed, while Susuman shrank from 17k to 4,3k and Magadan city from 152k to 90k (-40,8%) [31][34]. In Region 31 (Norilsk, Taymyr and Evenkia, declining from 380k to 220k, or -42,1%), the nickel centre of Norilsk contracted from 265k to 175k (-34%), Dudinka from 32k to 19,6k, the timber port of Igarka by -80,7% (18,8k to 3,6k) and Tura in Evenkia from 7,5k to 4,4k. Similarly, Region 15 (Murmansk, Arkhangelsk, Komi and Nenets AO) lost 1,56M residents (-36,1%, 4,32M to 2,76M), led by Vorkuta (116k to 57k, -50,9%), Inta (60k to 20k, -66,7%), Murmansk (468k to 270k, -42,3%), Apatity (88k to 50k), Arkhangelsk (416k to 301k) and Severodvinsk (249k to 157k) [32][35].

In the indigenous urbanisation and high-fertility southern Siberian republics (Regions 33, 29 and 32), Region 33 (Sakha Republic/Yakutia, shifting from 1,11M in 1991 to 950k in 2005 and 985k in 2021, or -11,3%) saw the departure of more than 250k ethnic Russians from northern gold, tin and coal towns (Tiksi 11,6k to 4,2k, Neryungri 82k to 57k and Ust-Nera) largely balanced by higher rural Sakha fertility and migration into Yakutsk, which nearly doubled from 187k in 1989 to 355k in 2021 (+89,8%) [14][33]. Region 29 (Altai, Khakassia and Tuva) held almost steady (1,09M to 1,08M, -0,5%): in Tuva, where ethnic Russian out-migration during the early 1990s reduced the Russian share from 32% in 1989 to 10,1% in 2021, Russia's highest total fertility rate (~2,7-3) lifted Kyzyl by +43,7% (87k to 125k), alongside Gorno-Altaysk (47k to 65k, +39,3%) and Abakan (156k to 185k) [5]. In Region 32 (Buryatia and Zabaykalsky Krai, 2,41M to 2M, -17,2%), Buryat rural-to-urban migration expanded Ulan-Ude from 353k to 438k (+24,1%) even as Zabaykalsky Krai (Chita and Krasnokamensk) contracted.

Across the Arctic hydrocarbon zone and the Trans-Siberian corridors (Regions 26, 27, 28, 30 and 35), oil and gas extraction in Region 26 (Khanty-Mansi/Yugra and Yamalo-Nenets autonomous okrugs) lifted the population by +23,9% (1,82M to 2,25M), led by Surgut (248k to 396k, +59,7%), Nizhnevartovsk (245k to 283k), Novy Urengoy (95k to 107k), Noyabrsk (100k), Khanty-Mansiysk (34k to 107k, +214,7%) and Salekhard (32k to 48k). Along the West Siberian Corridor (Region 27: 11,15M to 10,49M, -5,9%), rural contraction in Omsk and Altai Krai (Rubtsovsk and Biysk) was cushioned by the expansion of Tyumen (477k to 847k, +77,6%), Novosibirsk (1,44M to 1,63M) and the university city of Tomsk (506k to 556k). By

contrast, mine closures in Region 28 (Kuzbass/Kemerovo Oblast) reduced the regional total by -18,2% (3,18M to 2,6M; Prokopyevsk 274k to 178k, Kiselevsk 128k to 83k, Novokuznetsk 600k to 537k) [35]. Further east, Region 30 (Krasnoyarsk and Irkutsk) fell by -14,3% (6,03M to 5,17M) even as Krasnoyarsk city passed the million mark (913k to 1,19M, +30,1%), while Region 35 (Southern Far East: Primorye, Khabarovsk, Amur, Jewish AO and Sakhalin) declined by -24,3% (6,06M to 4,58M), with Komsomolsk-on-Amur shrinking from 318k to 238k (-25,2%) while Khabarovsk (601k to 617k) and Vladivostok (634k to 601k) drew in young migrants from the surrounding countryside [33][36].

Industrial Town Shrinkage and Metropolitan Growth

In European Russia, Ukraine and Belarus, Natalya Zubarevich's "Four Russias" model captures the divergence between shrinking industrial and rural peripheries and expanding metropolitan cores [24][26][35][54].

Across the interior belts (Regions 14, 16-17, 19-20 and 25) - including the Northwest and Karelia (Region 14, -24,7%), the Central Non-Black-Earth Ring (Region 16, -21,1%) and the Middle Volga and Finno-Ugric republics (Region 19, -13,5%) - older industrial centres contracted steadily (Ivanovo 481k to 362k, Tula 540k to 473k, Rybinsk 251k to 177k, Dzerzhinsk 285k to 219k, Nizhny Novgorod 1,44M to 1,23M), while rural contraction (*szhatie prostranstva*) left more than 19,4k Russian villages uninhabited (*derevni bez naseleniya*) by 2021 [5][55]. In the Urals (Region 25: 16,32M to 15,7M, -3,8%), the expansion of Yekaterinburg (1,37M to 1,54M) and Chelyabinsk (1,15M to 1,19M) contrasted with steep declines in metallurgical and potash *monogoroda* (Nizhny Tagil -23%, Zlatoust -22,1%, Berezniki -31,3%, Orsk -30,8%).

Conversely, metropolitan concentration reshaped the major urban regions (Regions 12, 13, 21, 18, 1 and 5). Region 12 (Moscow City and Moscow Oblast) grew by +5,35M (+33,9%, 15,8M to 21,15M), concentrating 14,4% of the Russian Federation's population within Moscow city (8,97M to 13,01M) and its commuter belt (Balashikha 135k to 521k [+285,9%], Podolsk 315k, Khimki 257k, Mytishchi 255k, Korolev 228k, Lyubertsy 224k, Krasnogorsk 188k, Odintsovo 181k and Domodedovo 152k) [27][52]. Region 13 (Saint Petersburg, Leningrad and Kaliningrad) rose by +10,5% (7,62M to 8,42M; Saint Petersburg 5,6M, suburb Murino 5k to 89k, Kaliningrad 490k), Region 21 (Southern Sunbelt) by +11,9% (13,15M to 14,71M; Krasnodar 621k to 1,1M [+77%], Sochi 466k, Stavropol 547k, Rostov-on-Don 1,14M), Region 18 (Tatarstan and Bashkortostan) by +4,9% (7,58M to 7,95M; Kazan 1,31M, Ufa 1,15M), Region 1 (Minsk) by +2,4% (3,4M to 3,48M; Minsk 1,61M to 2,01M) and Region 5 (Kyiv) by +4,6% (4,58M to 4,79M; Kyiv 2,59M to 2,96M, Brovary 109k, Irpin 62k, Bucha 37k).

The Chechen Wars and North Caucasus Demography

In sharp contrast to the ageing East Slavic interior, the three North Caucasus regions - Chechnya and Ingushetia (Region 22), North Ossetia-Alania, Kabardino-Balkaria and Karachay-Cherkessia (Region 23) and Dagestan (Region 24) - experienced both the most destructive armed conflicts on Russian territory since 1945 and the fastest long-term population growth across the 35-region division (+54,5% in Region 22, +9,9% in Region 23 and +77,1% in Region 24) [28][29][30]. Table 4 summarises the principal post-Soviet armed conflicts, casualties and forced migration flows across both the North Caucasus (Regions 22-24, 1992-2009) and Ukraine (Regions 9-11, 2014-2021).

Conflict and Region	Principal Settlements Affected	Estimated Deaths	Displacement and Population Shift
East Prigorodny Conflict (Region 23 North Ossetia / Region 22 Ingushetia, Oct-Nov 1992)	Vladikavkaz suburbs (Kambileev-skoye, Chermen, Kurtat), Beslan; Nazran and Sunzha camps	600 killed (plus 334 killed in Sep 2004 Beslan school siege)	40k-60k ethnic Ingush displaced from Prigorodny District to Ingushetia; 35k South Ossetian refugees into Reg. 23
First Chechen War (Dec 1994 - Aug 1996) and Interwar Ichkeria (Region 22)	Grozny (400k in 1989 to <90k in Feb 1995 siege); Gudermes, Argun, Shali, Samashki	35k-50k+ military and civilian deaths	Region 22 fell from 1,32M (1991) to 1,18M (1995); permanent exodus of >220k ethnic Russians from Chechnya
Second Chechen War and Counter-Insurgency (Aug 1999 - Apr 2009, Region 22)	Grozny (reduced to 50k-60k in Jan 2000; rebuilt to 211k by 2002, 272k by 2010); Urus-Martan	25k-40k+ military and civilian deaths	Region 22 bottomed at 1,12M (2000) before rising to 2,04M (2021, +54,5%); 350k Chechen IDPs fled to Ingushetia (Nazran 18k to 125k)
Dagestan Demographic Transition and Urban Boom (Region 24, 1991-2021)	Makhachkala (318k to 623k, +95,9%); Khasavyurt (70k to 155k); Derbent (78k to 125k); Kaspiysk (60k to 121k)	1999 Botlikh/Kadar incursion and low-level insurgency	Region 24 surged from 1,88M (1991) to 3,33M (2021, +77,1%) via high natural increase and mountain-to-piedmont migration
Annexation of Crimea and Sevastopol (Region 11, Feb-Mar 2014 - 2021)	Sevastopol (344k in 2014 to 548k in 2021 Census); Simferopol (341k); Kerch (155k); Yevpatoria (108k)	<10 direct deaths during Mar 2014 takeover	Region 11: 60k-100k Ukrainian and Tatar IDPs out vs 250k-800k Russian settler and military inflow (2,34M to 2,44M)
War in Donbas: Partition of Region 9 (GCA) vs Region 10 (Occupied Donbas, 2014-2021)	427 km Contact Line: Reg. 10 Occupied (Donetsk 905k, Luhansk 399k, Makiivka, Horlivka) vs Reg. 9 Ukrainian-Held (Mariupol 431k, Kramatorsk 150k, Sievierodo-netsk 101k)	14,4k killed (3,4k+ civilians, 4,4k Ukrainian forces, 6,5k separatist/ Russian forces)	Reg. 9 fell from 2,86M (2013) to 2,21M (2021); Reg. 10 fell from 3,59M (2013) to 2,47M (2021); 1,46M IDPs to Regs 5/7 and 1M to Russia

Table 4. Principal armed conflicts, territorial partitions, estimated war deaths and forced migration flows in the North Caucasus and Ukraine between 1992 and 2021.

In Region 22 (Chechnya and Ingushetia, which fell from 1,32M in 1991 to 1,12M in 2000 before rising to 2,04M in 2021, or +54,5%), the unified Checheno-Ingush ASSR at the 1989 Soviet census held 1,27M inhabitants (1,32M in 1991), including 735k Chechens, 164k Ingush and 294k Russians (23,1%), who were concentrated in Grozny (399,6k) [1][29]. During the First (1994-1996) and Second (1999-2009) Chechen Wars, siege warfare devastated Grozny (reduced to fewer than 60k residents in early 2000), Gudermes, Argun, Shali and Urus-Martan, driving Region 22 down to 1,18M in 1995 and 1,12M in 2000 [29][30]. More than 320k-350k displaced Chechens crossed into Ingushetia, joining 40k-60k Ingush expelled earlier from Prigorodny District in 1992 and causing Nazran to swell nearly sevenfold from 18,2k in 1989 to 125,1k at

the 2002 census (122k in 2021 alongside the new republic capital of Magas at 15k) [8][30]. Over 250k ethnic Russians permanently left Chechnya and Ingushetia for the Southern Sunbelt (Region 21), reducing the Russian share in Chechnya to 1,2% by 2021. Post-2004 federal reconstruction rebuilt Grozny (from 211k in 2002 to 272k in 2010 and 329k in 2021), while high Vainakh fertility (TFR ~2,5-2,8) and a young age structure (median age ~24 years) lifted Region 22 to 2,04M (1,51M in Chechnya and 510k in Ingushetia at the 2021 census) [5].

In Region 23 (North Ossetia-Alania, Kabardino-Balkaria and Karachay-Cherkessia, growing from 1,92M to 2,11M, or +9,9%), the East Prigorodny conflict outside Vladikavkaz in October-November 1992 killed roughly 600 people and displaced 40k-60k ethnic Ingush from Prigorodny District into Ingushetia (Region 22), while North Ossetia (Vladikavkaz 300k to 296k, and Beslan 33k to 36k, site of the September 2004 school siege) absorbed more than 35k Ossetian refugees from the 1991-1992 war in South Ossetia [28][30]. Across Kabardino-Balkaria (Nalchik 235k to 247k, Prokhladny 58k to 60k) and Karachay-Cherkessia (Cherkessk holding at 113k), indigenous natural increase offset gradual Russian out-migration, stabilising Region 23 at 2,11M between 2005 and 2021.

In Region 24 (Dagestan, expanding from 1,88M to 3,33M, or +77,1%), the population nearly doubled between 1991 and 2021 (+1,45M), driven by sustained natural increase (TFR ~1,9-2,4 paired with low adult mortality, CDR ~5,5-7 per 1k) and large-scale internal resettlement of Avar, Dargin, Lezgin, Kumyk and Lak mountain villagers to the Caspian coastal plain [5][28]. Makhachkala nearly doubled from 318k in 1989 to 623k in 2021 (+95,9%, or more than 1M across its wider agglomeration), while Khasavyurt rose by +121,4% (70k to 155k), Kaspiysk by +101,7% (60k to 121k) and Derbent by +60,3% (78k to 125k).

Crimea, the Donbas War and the COVID-19 Pandemic

The final seven years of the 1991-2021 period were shaped by the annexation of Crimea (Region 11) and the war and partition of the Donbas (Regions 9 and 10) from 2014 onward, followed by the COVID-19 mortality wave of 2020-2021 [17][25][56][57].

Annexation of Crimea and Sevastopol

Following the Russian annexation of the Autonomous Republic of Crimea and Sevastopol (Region 11) in February-March 2014, the peninsula's population shifted from 2,34M in 2013 to 2,36M in 2015 and 2,44M in the 2021 benchmark (2,48M in the October 2021 Russian census) [5][10]. Between 60k and 100k Ukrainian citizens and Crimean Tatars relocated to mainland Ukraine (Regions 4, 5 and 8), whereas 250k-350k+ permanent Russian military personnel, federal civil servants and retirees (and up to 500k-800k cumulative gross arrivals including construction workers and temporary residents between 2014 and 2021) moved onto the peninsula, lifting Sevastopol from 344k in 2014 to 548k in 2021 alongside Simferopol (341k), Kerch (155k) and Yevpatoria (108k) [17][50].

The War in Donbas

In April 2014, armed conflict broke out across Donetsk and Luhansk oblasts (6,45M combined residents in 2013), dividing the Donbas coal-and-steel conurbation along a fortified 427 km Line of Contact (Minsk II) into two demographic zones [3][17][25].

Region 10 (Pre-2022 Occupied Donbas, which fell from 4,6M in 1991 to 3,59M in 2013 and 2,47M in 2021, or -46,3%) encompassed the occupied urban core of Donetsk (965k in 2013 to ~905k nominal or fewer than 850k de facto in 2021), Luhansk (426k to ~399k), Makiivka (353k to ~335k), Horlivka (256k to ~235k), Alchevsk (111k to 105k), Yenakiieve (82k to 75k) and Kadiivka/Stakhanov (77k to 72k). Between 2013 and 2015 alone, Region 10 lost 890k inhabitants (-24,8%) through mass flight, followed by steep natural decrease (CDR > 21 per 1k vs CBR < 5,5 per 1k) that reduced its population to 2,47M by 2021 [17].

Region 9 (Government-Controlled Donbas, declining from 3,62M in 1991 to 2,86M in 2013 and 2,21M in 2021, or -39%) covered Ukrainian-held Mariupol (459k to 431k), the provisional Donetsk oblast seat of Kramatorsk (163k to 150k), Sloviansk (117k to 106k), the provisional Luhansk oblast seat of Sievierodonetsk (109k to 101k), Lysychansk (104k to 93k), Kostiantynivka (76k to 68k), Bakhmut (77k to 71k) and Pokrovsk (65k to 60k).

Up to 31 December 2021, the war in the Donbas killed roughly 14,4k people and displaced more than 2,5M residents. About 1,46M registered internally displaced persons relocated within government-controlled Ukraine - above all into neighbouring Region 7 (Kharkiv, Dnipropetrovsk and Zaporizhzhia, whose population rose briefly from 7,62M in 2013 to 7,65M in 2015) and Region 5 (Kyiv, rising from 4,58M to 4,66M in 2015) - while roughly 1M refugees and migrants moved into the Russian Federation (Rostov and Krasnodar in Region 21, Voronezh and Belgorod in Region 17 and Moscow in Region 12) and Belarus (Regions 1-3) [17][18][25].

Excess Mortality from COVID-19

After a brief window of natural increase in Russia in 2013-2015 (+83k), structural birth decline returned after 2016 as the small cohorts born in the 1990s reached childbearing age, followed in 2020-2021 by the COVID-19 pandemic [40][56][57][58].

Across the Russian Federation (Regions 12-35), registered deaths surged from 1,8M in 2019 to 2,14M in 2020 and 2,44M in 2021 (16,7 per 1k), producing 1,07M-1,15M cumulative excess deaths, pushing annual natural decline to a post-1945 record of -1,04M in 2021 alone and reducing Russia's total population across Regions 12-35 from 148,07M in 2018 to 147,37M in 2021 [57][58].

In Ukraine (Regions 4-11), roughly 185k COVID-19 excess deaths pushed annual natural decline to -442k in 2021 (CDR = 18,5 per 1k vs CBR = 7,3 per 1k), reducing Regions 4-11 from 42,33M in 2018 to 41,56M in 2021 [3][18]. In Belarus (Regions 1-3), 70k-85k excess deaths and post-2020 emigration reduced the population from 9,25M in 2018 to 9,15M in 2021, bringing the 35-region post-Soviet space to 198,08M on the eve of the February 2022 invasion [4][56].

Sources

1. Goskomstat SSSR. *Itogi Vsesoyuznoy Perepisi Naseleniya 1989 Goda: Chislennost i Razmeshchenie Naseleniya SSSR*. Finansy i statistika, 1990, https://www.demoscope.ru/weekly/ssp/census.php.
2. Rosstat. *Demograficheskiy Ezhegodnik Rossii 2022: Statisticheskiy Sbornik*. Rosstat, 2022, https://rosstat.gov.ru/folder/12781.
3. Derzhstat Ukrainy. *Naselennia Ukrainy Za 2020 Rik: Demohrafichnyi Shchorichnyk; Chiselnist Nayavnoho Naselennia Ukrainy Na 1 Sichnia 2021 Roku*. Derzhavna sluzhba statystyky Ukrainy, 2021, https://www.ukrstat.gov.ua/.
4. Belstat. *Itogi Perepisi Naseleniya Respubliki Belarus 2019 Goda: V 7 Tomakh*. Natsionalnyy statisticheskiy komitet Respubliki Belarus, 2020, https://www.belstat.gov.by/.
5. Rosstat. *Itogi Vserossiyskoy Perepisi Naseleniya 2020 Goda (Po Sostoyaniyu Na 1 Oktyabrya 2021 Goda). Tom 1: Chislennost i Razmeshchenie Naseleniya*. Rosstat, 2022, https://rosstat.gov.ru/folder/12781.
6. Dubilet, Dmytro, et al. *Otsinka Chyselnosti Nayavnoho Naselennia Ukrainy Stanom Na 1 Hrudnia 2019 Roku (Electronic Population Assessment)*. Cabinet of Ministers of Ukraine; Ptoukha Institute for Demography; Social Studies, 2020, https://idss.org.ua.
7. Clem, Ralph S. *Research Guide to the Russian and Soviet Censuses*. Cornell University Press, 2016, https://doi.org/10.7591/9781501707087.

8. Rosstat. *Itogi Vserossiyskoy Perepisi Naseleniya 2002 Goda: V 14 Tomakh. Tom 1: Chislennost i Razmeshchenie Naseleniya*. IIC Statistika Rossii, 2004, https://rosstat.gov.ru/folder/12781.
9. Rosstat. *Itogi Vserossiyskoy Perepisi Naseleniya 2010 Goda: V 11 Tomakh. Tom 1: Chislennost i Razmeshchenie Naseleniya*. IIC Statistika Rossii, 2012, https://rosstat.gov.ru/folder/12781.
10. Rosstat. *Itogi Perepisi Naseleniya v Krymskom Federalnom Okruge 2014 Goda*. Rosstat, 2015, https://rosstat.gov.ru/folder/12781.
11. Vishnevsky, Anatoly G. *Demograficheskaya Modernizatsiya Rossii, 1900–2000*. Novoe izdatelstvo, 2006, https://search.worldcat.org/search?q=Demograficheskaya+modernizatsiya+Rossii+1900+2000+Vishnevsky+2006.
12. Zakharov, Sergei V. "Russian Federation: From the First to Second Demographic Transition." *Demographic Research*, vol. 19, 2008, pp. 907-72, https://doi.org/10.4054/DemRes.2008.19.24.
13. Andreev, Evgueni M., et al. "Health Expectancy in the Russian Federation: A New Perspective on the Health Divide in Europe." *Bulletin of the World Health Organization*, vol. 81, no. 11, 2003, pp. 778-87, https://pmc.ncbi.nlm.nih.gov/articles/PMC2572496.
14. Smirnov, Andrey V. "Migration Landscape of Russia: Typology of Municipalities and Patterns of Spatial Population Mobility." *Population and Economics*, vol. 10, no. 1, 2026, pp. 1-28, https://doi.org/10.3897/popecon.10.e146614.
15. Derzhkomstat Ukrainy. *Vseukrainskyi Perepys Naselennia 2001 Roku: Kilkist Ta Terytorialne Rozmishchennia Naselennia Ukrainy*. Derzhkomstat Ukrainy, 2003, https://www.ukrstat.gov.ua/.
16. Libanova, Ella M., et al. *Naselennia Ukrainy: Imperatyvy Demohrafichnoho Starinnia Ta Depopuliatsii*. Ptoukha Institute for Demography; Social Studies, NAS of Ukraine, 2014, https://search.worldcat.org/search?q=Naselennia+Ukrainy+Imperatyvy+demohrafichnoho+starinnia+ta+depopuliatsii+Libanova+2014.
17. Gladun, Oleksandr M. "Demographic Losses and Migration Shifts in Ukraine Under Conditions of Armed Conflict in Donbas (2014–2020)." *Demography and Social Economy*, vol. 40, no. 2, 2020, pp. 3-24, https://dse.org.ua/en/archive.
18. Perelli-Harris, Brienna, and Yuliya Hilevych. "The Triple Burden of Depopulation in Ukraine: Examining Perceptions of Population Decline." *Vienna Yearbook of Population Research*, vol. 21, 2023, pp. 1-28, https://doi.org/10.1553/p-azcj-d4f4.
19. Minstat Respubliki Belarus. *Itogi Perepisi Naseleniya Respubliki Belarus 1999 Goda*. Minstat RB, 2000, https://www.belstat.gov.by/.
20. Belstat. *Perepis Naseleniya Respubliki Belarus 2009 Goda: Chislennost, Sostav i Razmeshchenie Naseleniya*. Natsionalnyy statisticheskiy komitet Respubliki Belarus, 2010, https://www.belstat.gov.by/.
21. Shakhotska, Lyudmila P. *Demograficheskiy Potentsial Respubliki Belarus: Analiz i Perspektivy Razvitiya*. Belorusskaya nauka, 2009, https://search.worldcat.org/search?q=Demograficheskiy+potentsial+Respubliki+Belarus+Analiz+perspektivy+razvitiya+Shakhotska+2009.
22. Antipova, Ekaterina A. *Geografiya Selskogo Rasseleniya Belarusi: Prostranstvenno-Vremennie Zakonomernosti*. BGU, 2013, https://search.worldcat.org/search?q=Geografiya+selskogo+rasseleniya+Belarusi+Prostranstvenno+vremennie+zakonomernosti+Antipova+2013.
23. Silvagni, Maria Giulia. *Patterns of Regional Differentiation in Post-Soviet Russia*. University of Bologna, 2012, https://scholar.google.com/scholar?q=Patterns+Regional+Differentiation+Post+Soviet+Russia+Silvagni+2012.
24. Zubarevich, Natalya V. "Four Russias: Human Potential and Social Differentiation of Russian Regions and Cities." *Regional Research of Russia*, vol. 3, no. 4, 2013, pp. 321-34, https://doi.org/10.1057/9781137336910_4.
25. UNHCR Ukraine, and Ministry of Social Policy of Ukraine. *Registration of Internal Displacement and*

Conflict-Affected Population Along the Donbas Contact Line (2014–2021). United Nations High Commissioner for Refugees, 2021, https://data.unhcr.org/en/situations/ukraine.

26. Karachurina, Liliya B., and Nikita V. Mkrtchyan. "The Role of Migration in Enhancing Settlement Pattern Contrasts at the Municipal Level in Russia." *Regional Research of Russia*, vol. 8, no. 3, 2018, pp. 258-70, https://doi.org/10.1134/s2079970516040080.
27. Mkrtchyan, Nikita V. "Migration in the Moscow Agglomeration: Suburban Boom and Core Polarisation." *Regional Research of Russia*, vol. 10, no. 2, 2020, pp. 192-204, https://doi.org/10.1134/S2079970520020094.
28. Belozerov, Vitaly S. *Etnodemograficheskie Protsessy Na Severnom Kavkaze*. Izdatelstvo SGU, 2005, https://search.worldcat.org/search?q=Etnodemograficheskie+protsessy+Severnom+Kavkaze+Belozerov+2005.
29. Tishkov, Valery A. *Chechnya: Life in a War-Torn Society*. University of California Press, 2004, https://doi.org/10.1525/9780520930209.
30. Vendina, Olga I., et al. "The Wars in Chechnya and Their Effects on Neighboring Regions." *Eurasian Geography and Economics*, vol. 48, no. 2, 2007, pp. 178-201, https://doi.org/10.2747/1538-7216.48.2.178.
31. Heleniak, Timothy. "Out-Migration and Depopulation of the Russian North During the 1990s." *Post-Soviet Geography and Economics*, vol. 40, no. 3, 1999, pp. 155-205, https://doi.org/10.1080/10889388.1999.10641111.
32. Heleniak, Timothy. "The Role of Attachment to Place in Migration Decisions of the Population of the Russian North." *Polar Geography*, vol. 32, no. 1-2, 2009, pp. 31-60, https://doi.org/10.1080/10889370903000398.
33. Mkrtchyan, Nikita V. "Migration Inside Russia: The Western Drift and Centripetal Concentration." *Regional Research of Russia*, vol. 5, no. 4, 2015, pp. 340-52, https://doi.org/10.1134/S2079970515040115.
34. Hill, Fiona, and Clifford G. Gaddy. *The Siberian Curse: How Communist Planners Left Russia Out in the Cold*. Brookings Institution Press, 2003, https://search.worldcat.org/search?q=Siberian+Curse+How+Communist+Planners+Left+Russia+Out+Hill+2003.
35. Zubarevich, Natalya V. "Transformation of Russian Single-Industry Towns (Monogoroda) in the Post-Soviet Period." *Voprosy Ekonomiki*, no. 8, 2019, pp. 105-22, https://doi.org/10.32609/0042-8736-2017-7-46-61.
36. Contreras-Luna, Rafael. "Moscow's Great Power Dilemmas: The Role of Asiatic Russia and Far Eastern Depopulation." *International Journal of Asia Pacific Studies*, vol. 15, no. 2, 2019, pp. 1-29, https://doi.org/10.21315/ijaps2019.15.2.1.
37. Shkolnikov, Vladimir M., et al. "Mortality Reversal in Russia: The Story so Far." *Hygiea Internationalis*, vol. 4, no. 1, 2004, pp. 29-80, https://doi.org/10.3384/hygiea.1403-8668.044129.
38. Mesle, France, and Jacques Vallin. *Mortality and Causes of Death in 20th-Century Ukraine*. Springer, 2012, https://doi.org/10.1007/978-94-007-2433-4.
39. Billingsley, Sunnee. *Casualties of Turbulent Economic Transition: Premature Mortality and Foregone Fertility in the Post-Communist Countries*. Universitat Pompeu Fabra, 2009, https://scholar.google.com/scholar?q=Casualties+Turbulent+Economic+Transition+Premature+Mortality+Foregone+Fertility+Billingsley+2009.
40. Yastrebov, Gordey A. "The Demographic Echo of War and Educational Attainment in Soviet and Post-Soviet Russia." *Demographic Research*, vol. 45, 2021, pp. 731-64, https://doi.org/10.4054/DemRes.2021.45.22.
41. Bhattacharya, Jay, et al. *The Gorbachev Anti-Alcohol Campaign and Russia's Mortality Crisis*. National Bureau of Economic Research, 2012, https://doi.org/10.3386/w18589.
42. Leon, David A., et al. "Hazardous Alcohol Drinking and Premature Mortality in Russia: A Population

Based Case-Control Study." *The Lancet*, vol. 369, no. 9578, 2007, pp. 2001-09, https://doi.org/10.1016/s0140-6736(07)60941-6.

43. Brainerd, Elizabeth. "Economic Reform and Mortality in the Former Soviet Union: A Study of the Suicide Epidemic in the 1990s." *European Economic Review*, vol. 45, no. 4-6, 2001, pp. 1007-19, https://doi.org/10.2139/ssrn.265620.
44. Shkolnikov, Vladimir M., et al. "Components and Possible Determinants of the Decrease in Russian Mortality in 2004–2010." *Demographic Research*, vol. 28, 2013, pp. 917-50, https://doi.org/10.4054/DemRes.2013.28.32.
45. Rudnytskyi, Omelian, et al. "Demography of a Man-Made Human Catastrophe: The Case of Massive Famine and Long-Term Demographic Shrinkage in Ukraine." *Canadian Studies in Population*, vol. 42, no. 1-2, 2015, pp. 53-80, https://doi.org/10.15407/dse2015.03.003.
46. Locher, Lilo. *Migration in the Soviet Successor States*. Institute for the Study of Labor (IZA), 2002, https://doi.org/10.2139/ssrn.341560.
47. Karacay, Aysem Biriz. "Contextualizing Russian Diaspora and Post-Soviet Emigration Trajectories." *International Journal of Social Sciences*, vol. 9, no. 37, 2025, pp. 1-22, https://doi.org/10.52096/usbd.9.37.19.
48. Abylkalikov, Salavat I. "The Waves of Internal Migration in Russia." *Academic Journal of Interdisciplinary Studies*, vol. 2, no. 8, 2013, pp. 228-32, https://doi.org/10.5901/ajis.2013.v2n9p572.
49. Smutchak, Zinaida. "Consequences of Demographic Collapse in Ukraine." *Innovative Economics and Management*, vol. 10, no. 2, 2023, pp. 45-54, https://doi.org/10.46361/2449-2604.10.3.2023.14-23.
50. Burdonosova, Maryna A. "Demographic Crisis and Population Displacement in Post-1991 Ukraine." *Knowledge Education Law Management*, vol. 2, 2026, pp. 112-25, https://doi.org/10.51647/kelm.2026.2.19.
51. Kauffmann, Albrecht. *The Russian Urban System in Transition: The View of New Economic Geography*. European Regional Science Association (ERSA), 2013, https://www.ersa.org.
52. Makhrova, Alla G., et al. "Moscow Agglomeration: Urbanisation, Suburbanisation, and Post-Suburban Seasonal Dacha Mobility." *Regional Research of Russia*, vol. 6, no. 4, 2016, pp. 332-44, https://doi.org/10.1134/S2079970516010081.
53. Slepukhina, Irina. *Regional Policy and Urbanization in Contemporary Russia*. European Regional Science Association (ERSA), 2012, https://www.ersa.org.
54. Zubarevich, Natalya V. *Regiony Rossii: Neravenstvo, Krizis, Modernizatsiya*. Nezavisimyy institut sotsialnoy politiki, 2010, https://search.worldcat.org/search?q=Regiony+Rossii+Neravenstvo+krizis+modernizatsiya+Zubarevich+2010.
55. Nefedova, Tatyana G., and Andrey I. Treivish. "Cities and Rural Countryside in Russia: Compression and Polarisation of Socio-Economic Space." *Regional Research of Russia*, vol. 1, no. 1, 2010, pp. 23-36, https://doi.org/10.1134/S2079970511010096.
56. Kobak, Dmitry. "Excess Mortality in Russia During the COVID-19 Pandemic in 2020–2021." *Significance*, vol. 18, no. 3, 2021, pp. 16-19, https://doi.org/10.5604/01.3001.0015.0351.
57. Timonin, Sergey, et al. "Excess Mortality in Russia and Its Regions Compared to High Income Countries During the COVID-19 Pandemic, 2020–2021." *Population Health Metrics*, vol. 20, no. 1, 2022, pp. 1-15, https://doi.org/10.1016/j.ssmph.2021.101006.
58. Scholey, Jonas, et al. "Life Expectancy Changes Since COVID-19." *Nature Human Behaviour*, vol. 6, 2022, pp. 1649-59, https://doi.org/10.1038/s41562-022-01450-3.